

1. ~~Overfitting~~ Underfitting: A model is too simple to capture pattern in data
→ Train and test set error are high

Overfitting: A model is too complex and learn noise from data
→ Train error low, test error high

2. Bias-Variance decomposition is $E[(y - \hat{f}(y))^2] = \text{Bias}^2 + \text{Var} + \sigma^2$.

Bias² = error from too simple model (Underfit)

Variance = how much model's output change across training samples (overfit)

σ^2 = irreducible noise in data

3. Low train error, high validation error → high variance → Overfit

Two ways to reduce it: - Regularization

└ Reducing model complexity, more data

4. Random row split can cause leakage because window from same subject is related → can appear in train and test sets.

→ Model may learn subject-specific pattern rather than general patterns.

5. At x , $\text{Var}(\hat{f}(x)) = \frac{1}{3} \sum_i (\hat{f}_i(x) - E(\hat{f}(x)))^2$ with $E(\hat{f}(x)) = \frac{1}{3} (\hat{f}_1 + \hat{f}_2 + \hat{f}_3)$. If $\hat{f}_1 = \hat{f}_2 = \hat{f}_3 \rightarrow \text{Var} = 0$.

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1. Entropy : $H(S) = -\sum_i p_i \log_2(p_i)$ It measures how mixed classes are.

Entropy = 0 $\rightarrow$ all sample belongs to 1 class.

For binary classification, It is 1 when 2 classes are evenly split, $p=0.5$ for each.

2. Information Gain is how much a split reduces the entropy of a node.

$IG(S, A) = H(S) - \sum_v \left(\frac{|S_v|}{|S|} \right) H(S_v)$. ID3 calculates the IG for each candidate attribute and choose the attribute with highest value to split node.

3. ID3 doesn't directly handle continuous attribute or missing values $\rightarrow$ preprocess first.

C4.5 handles by searching for a threshold and handle missing values by distributing weight across possible branches.

CART: Binary threshold split and can use surrogated splits when primary feature is missing.

4. Pruning reduces overfitting by preventing the tree from being too complex.

- Pre-pruning: Stop growing when reach condition

- Post-pruning: Remove branches not providing much improvement for a tree.

$$5. H(S) = -\frac{6}{10} \log_2 \frac{6}{10} - \frac{4}{10} \log_2 \frac{4}{10} \approx 0.971$$

$$S_1 (\text{pure A}) : H(S_1) = 0$$

$$S_2 (1A, 4B) : H(S_2) = -\frac{1}{5} \log_2 \frac{1}{5} - \frac{4}{5} \log_2 \frac{4}{5} \approx 0.722$$

$$\rightarrow IG(S, X) = 0.971 - \left(\frac{5}{10}\right) \cdot 0 - \left(\frac{5}{10}\right) \cdot 0.722 \approx 0.610$$